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Double Layered Microshells Composed of Calcium Phosphate and Poly (Lactic Acid)

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Abstract. Biodegradable microshells were synthesized by an improvement surfactant-free emulsion solvent evaporation method. The oil in water emulsions composed of the organic phase of poly (lactic acid) in dichloromethane and the aqueous phase including Ca^{2+} ions and PO_4^{3-} ions were firstly cooled down by an ice bath to stabilize emulsification. During the emulsions was returning to room temperature, calcium phosphate was rapidly precipitated at the oil/water interface, which would result in stabilizing the microshell structure on unstable liquid oil droplets. The precipitated hard calcium phosphate shells acted as a framework for poly (lactic acid) deposition by volatilization of dicloromethane. The resulting microshells were approximately 50 - 300 μm in diameter with the wall thickness of 1 - 10 μm. The walls of the microshells had a double-layered structure composed of external walls of calcium phosphate and internal walls of poly (lactic acid). The biodegradable double-layered microshells were expected to be utilized as drug delivery carriers for injectable bone substitute.

Introduction

Calcium phosphate is one of the ideal materials to synthesize bone substitutes because of its biocompatibility and biodegradability. Microspheres with shell structures have possible applications such as therapeutic agents filled capsules for injectable bone substitutes. Previous studies have shown that the preparation of hollow microspheres involves various templating approaches using vesicles and emulsion droplets to prepare silica [1,2], titania [3], calcium carbonate [4] and calcium phosphate [5,6] microshells. Since most of these approaches use cytotoxic or non-biodegradable materials such as surfactants, they might possess biological problems for in vivo applications. Our previous work demonstrated that it was possible to prepare calcium phosphate/poly (lactic acid) microspheres with a surfactant-free emulsification method [7,8]. In this present paper, we describe a method for direct synthesis of double-layered microshell composed of calcium phosphate and poly (lactic acid) by a modified surfactant-free emulsification process.

Materials and Methods

Poly (lactic acid) (PLA, Mw = 20000) was purchased from Wako (Tokyo, Japan). All other chemicals were of analytical grade.

Modified oil-in-water emulsion solvent evaporation method was used for calcium phosphate / poly (lactic acid) microshells fabrication. The organic phase composed of PLA and dichloromethane was poured in calcium acetate ($(CH_3COOH)_2Ca$) aqueous solution. At first, the mixture was stirred at 300

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rpm to yield emulsion at room temperature, then the beaker filled with the emulsion was set in an ice bath or a cool bath (10 °C) to cool down for 30 min with stirring. Diammonium hydrogenphosphate ($(NH_4)_2HPO_4$) aqueous solution was slowly added into the emulsion to precipitate calcium phosphate. After adding diammonium hydrogenphosphate aqueous solution, the beaker was taken out from an ice bath or a cool bath and allowed to return to room temperature. The suspension was stirred for 24 h at room temperature until dichloromethane was removed thoroughly. In order to transform the crystal phase, some of the samples were aged for 120 h. The suspension was filtered, washed and freeze-dried to obtain solid phase.
The morphology of the products was observed using scanning electron microscope (SEM, HITACHI, S3000) and an optical microscope (Nikon, SZX-1500). Phase of the microspheres was analyzed by XRD (MAC science, MXP3).

Results and Discussion

The SEM images of the samples prepared in an ice bath are shown in Fig. 1. The images revealed that the products consisted of spherical particles with a wide distribution ranging from 50 to 300 μm. SEM image taken at higher magnifications showed that the surface of the particles was covered with many plate-like precipitates. Some of the particles were collapsed, which revealed the particles have a microshell structure with the wall thickness of 1-10 μm. The internal wall of the hollow was smooth, contrary to this, the external wall have rough morphology, which indicate that the wall would have multiple layers.

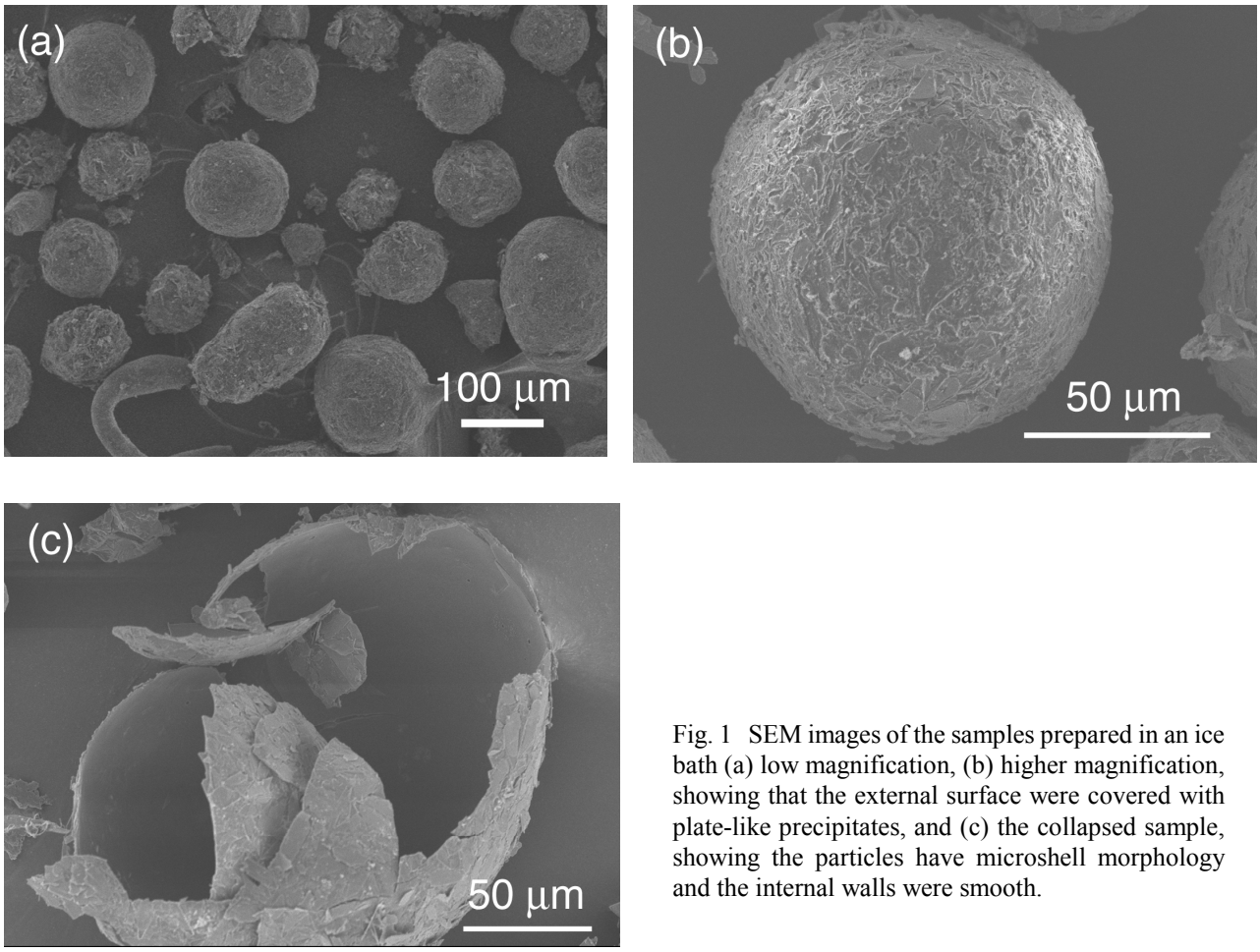

Fig. 1 SEM images of the samples prepared in an ice bath (a) low magnification, (b) higher magnification, showing that the external surface were covered with plate-like precipitates, and (c) the collapsed sample, showing the particles have microshell morphology and the internal walls were smooth.

EDX analysis confirmed the presence of the Ca and P elements on the external wall of the particles, showing the precipitate on the external wall to be calcium phosphate. The precipitates were identified as dicalcium phosphate dihydrate (DCPD) by XRD (Fig. 2(b)). Our previous study described that the crystal phase of the calcium phosphate precipitated on microspheres was low crystallized hydroxyapatite (HAp) at room temperature although the precipitate in water phase as homogeneous nucleation were DCPD [8]. It was explained that the arrange of the hydrophilic groups of PLA at the interface between oil and water phases would regulate the crystal phase under moderate calcium phosphate precipitations. In contrast, in this study, drastic change of the reacting temperature from 0 °C to room temperature would induce rapid precipitation which caused DCPD, the most stable phase around pH 6.5, on the oil droplets in the emulsion. For comparison, some samples were prepared in cool bath settled at 10 °C, indicating that the crystal phase was the mixture of HAp and DCPD (Fig. 2(a)), additionally the resulting particles were not sphere. The reacting temperature strongly affected the morphology of the particles as well as the crystal phase of the precipitate, that is, the moderate precipitated HAp would not be hard enough to maintain the microshell structure on liquid oil droplets during volatilization of dichloromethane. The rapidly precipitated DCPD caused by the temperature increase from 0 °C had an important role to stabilize the microshell structure on unstable liquid oil droplets. In order to transform the crystal phase of calcium phosphate from DCPD to HAp, some of the samples were aged for 120 h. Fig. 2(c) indicates that the crystal phase of the aged samples were transformed to HAp that is more useful for biomedical applications.

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Fig. 2 XRD patterns of the samples prepared in (a) cool bath (10 °C), (b) ice bath (0 °C), and (c) the long aged sample (120 h).

These results allowed us to assume as follows. An ice bath made the emulsion cold enough to reduce evaporation of dichloromethane in organic phase, stabilizing oil droplets in the oil-in-water emulsion. During the emulsion was returned to room temperature, DCPD was precipitated rapidly on the nucleation sites of dissolved poly (lactic acid) in the interface between water phase and oil droplets, which formed solid calcium phosphate, DCPD, microshell structure. Furthermore, with the reacting

temperature returning to room temperature, dichloromethane was volatilized, and so poly (lactic acid) would be deposited from the internal wall of DCPD microshell.

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Fig. 3 Schematic illustration of microshell formation.

Summary

In this paper, we have shown that it is possible to prepare double-layered micoshells composed of calcium phosphate and poly (lactic acid) by a surfactant-free emulsion method. Cooling down of the emulsion would play important role to control evaporation pressure of dichloromethane. In addition, the precipitated DCPD on the oil droplets would stabilize the microshell structure, which would cause double-layered microshell formation.

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Bioceramics 18
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